

Experiment 6

ARP & PPP Configuration

Lab 6 Objectives:

- Observe ARP message exchange and configure PPP with authentication for direct network connections.

Prerequisites: The prerequisites of this lab are-

- Basic networking knowledge (MAC & IP addressing, ARP, PPP).
- Familiarity with command-line tools for network configuration.

Outcomes: By the end of this lab, students will be able to:

- Capture and analyze ARP messages.
- Configure and verify PPP with authentication.
- Troubleshoot ARP and PPP-related issues.

What is ARP (Address Resolution Protocol)?

ARP is used to map IP addresses to MAC (hardware) addresses in a local area network. It is essential for devices to communicate within the same subnet.

How ARP Works:

1. If a device wants to send data to another device but only has its IP address, it sends an **ARP Request** (broadcast).
2. The target device replies with an **ARP Reply** (unicast), providing its MAC address.
3. The sender stores the MAC-IP mapping in its ARP cache.

The image shows a Wireshark packet capture of an ARP request and reply. The packet list on the left shows 14 packets. The packet details pane on the right shows the structure of the selected packet (packet 12, an ARP request).

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private 66:68:01	Broadcast	ARP	64	who has 10.0.0.1? Tell 10.0.0.2
2	0.000969	Private 66:68:00	Private 66:68:01	ARP	64	10.0.0.1 is at 00:50:79:66:68:00
3	0.002961	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) request id=0xb3eb, seq=1/256, ttl=64 (reply in 4)
4	0.004024	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) reply id=0xb3eb, seq=1/256, ttl=64 (request in 3)
5	1.008272	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) request id=0xb4eb, seq=2/512, ttl=64 (reply in 6)
6	1.008272	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) reply id=0xb4eb, seq=2/512, ttl=64 (request in 5)
7	2.061577	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) request id=0xb5eb, seq=3/768, ttl=64 (reply in 8)
8	2.062452	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) reply id=0xb5eb, seq=3/768, ttl=64 (request in 7)
9	3.063776	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) request id=0xb6eb, seq=4/1024, ttl=64 (reply in 10)
10	3.064772	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) reply id=0xb6eb, seq=4/1024, ttl=64 (request in 9)
11	4.066126	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) request id=0xb7eb, seq=5/1280, ttl=64 (reply in 12)
12	4.067091	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) reply id=0xb7eb, seq=5/1280, ttl=64 (request in 11)

Packet Details (Packet 12):

- Padding: 00000000000000000000000000000000
- Frame check sequence: 0x00000000 [unverified]
- [FCS Status: Unverified]
- Address Resolution Protocol (request)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcode: request (1)
 - Sender MAC address: Private 66:68:01 (00:50:79:66:68:01)
 - Sender IP address: 10.0.0.2
 - Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
 - Target IP address: 10.0.0.1

What is PPP (Point-to-Point Protocol)?

PPP is a data link layer protocol used to establish a direct connection between two networking nodes (usually routers) over serial links.

Features of PPP:

- Supports authentication using PAP or CHAP.
- Encapsulates network layer protocols.
- Error detection and compression.

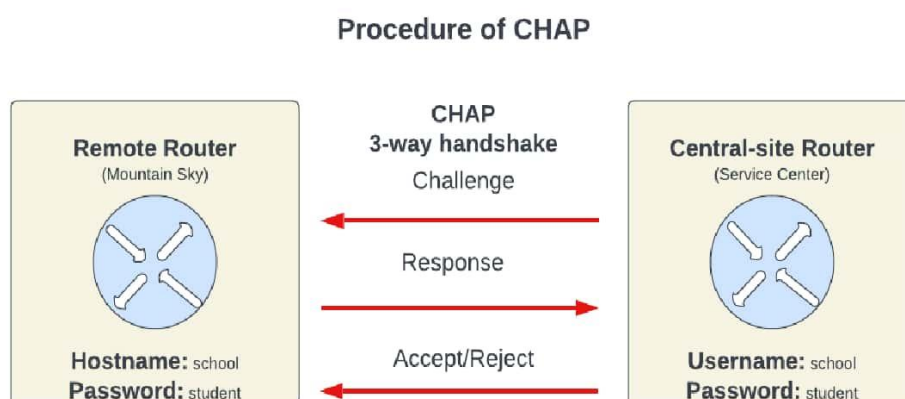
PPP Authentication Methods:

1. PAP (Password Authentication Protocol):

- Two-way handshake.
- Transmits username and password in plain text.
- Less secure.
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2. CHAP (Challenge Handshake Authentication Protocol):

- Three-way handshake.
- Encrypts the password using a hash function.
- More secure.



On **PC1 (my laptop)**, I used the ping command to initiate communication with another device in the same subnet:

```
C:\Users\CHETAN>ping 192.168.226.42

Pinging 192.168.226.42 with 32 bytes of data:
Reply from 192.168.226.42: bytes=32 time<1ms TTL=128
Reply from 192.168.226.42: bytes=32 time<1ms TTL=128
Reply from 192.168.226.42: bytes=32 time<1ms TTL=128
Reply from 192.168.226.42: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.226.42:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\CHETAN>
```

This confirms that the destination device is reachable, and the system successfully resolved the IP address using ARP.

Then, using **Wireshark**, I captured the ARP message exchange. I applied the filter:

arp					
Time	Source	Destination	Protocol	Length	No. Info
11.722458	4e:a9:95:a0:5c:c0	Intel_a0:10:13	ARP	42	1010 Who has 192.168.226.42? Tell 192.168.226.177
11.722488	Intel_a0:10:13	4e:a9:95:a0:5c:c0	ARP	42	1011 192.168.226.42 is at dc:45:46:a0:10:13
35.341694	4e:a9:95:a0:5c:c0	Intel_a0:10:13	ARP	42	1177 Who has 192.168.226.42? Tell 192.168.226.177
35.341723	Intel_a0:10:13	4e:a9:95:a0:5c:c0	ARP	42	1178 192.168.226.42 is at dc:45:46:a0:10:13
55.345069	4e:a9:95:a0:5c:c0	Intel_a0:10:13	ARP	42	1244 Who has 192.168.226.42? Tell 192.168.226.177
55.345100	Intel_a0:10:13	4e:a9:95:a0:5c:c0	ARP	42	1245 192.168.226.42 is at dc:45:46:a0:10:13
114.189117	4e:a9:95:a0:5c:c0	Intel_a0:10:13	ARP	42	2023 Who has 192.168.226.42? Tell 192.168.226.177
114.189153	Intel_a0:10:13	4e:a9:95:a0:5c:c0	ARP	42	2024 192.168.226.42 is at dc:45:46:a0:10:13

Frame 1010: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{7EC3D058-0000-0000-0000-000000000000} interface 0	dc 45 46 a0 10 13 4e a9 95 a0 5c c0 08 06 00 01	EF
Ethernet II, Src: 4e:a9:95:a0:5c:c0 (4e:a9:95:a0:5c:c0), Dst: Intel_a0:10:13 (dc:45:46:a0:10:13)	0010 00 00 06 04 00 01 4e a9 95 a0 5c c0 c0 a8 e2 b1	N
Address Resolution Protocol (request)	0020 00 00 00 00 00 c0 a8 e2 2a	*

Observation:

ARP Request: *Who has 192.168.226.42? Tell 192.168.226.177*

ARP Reply: *192.168.226.42 is at dc:65:56:a0:19:93*

Configure PPP Between Two Routers – Theory

Objective:

To configure a **Point-to-Point Protocol (PPP)** connection between two routers using serial interfaces for direct communication.

What is PPP?

Point-to-Point Protocol (PPP) is a **Data Link Layer (Layer 2)** protocol that is used to establish a **direct connection between two networking devices** such as routers over serial links. PPP supports:

- **Encapsulation of network layer protocols** (like IP, IPX)
- **Authentication mechanisms** (PAP and CHAP)
- **Compression and error detection**

Why Use PPP?

Although routers by default use **HDLC (High-Level Data Link Control)** as their encapsulation protocol on serial interfaces, PPP is more versatile and supports

- **Multivendor compatibility**
- **Advanced authentication (PAP/CHAP)**
- **Link quality monitoring**
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Therefore, PPP is widely used in WAN (Wide Area Network) links where authentication and reliability are important.

Required Setup:

To configure PPP, we need:

- Two routers with **serial interfaces**
- A **serial (DCE/DTE) cable** to connect the routers
- IP addresses on both interfaces that belong to the **same subnet**

Configuration Commands:

On Router1:

```
interface Serial0/0/0
ip address 10.0.0.1 255.255.255.252
encapsulation ppp
no shutdown
```

On Router2:

```
interface Serial0/0/0
ip address 10.0.0.2 255.255.255.252
encapsulation ppp
no shutdown
```

Explanation:

- interface Serial0/0/0: Enters the serial interface configuration mode.
- ip address: Assigns an IP address in the same subnet to the interface.
- encapsulation ppp: Changes the default encapsulation from HDLC to PPP.
- no shutdown: Activates the interface.

Expected Outcome:

After configuring PPP on both routers:

- The serial link becomes **active**.
- PPP is used for encapsulation.
- The routers can **communicate** with each other using IP over a **PPP link**.
- This setup can later be extended to add **authentication (CHAP/PAP)**.

Analysis Report

Experiment Title:

Observing ARP & Configuring PPP Authentication

The primary objective of this experiment was to:

- Understand the working of Address Resolution Protocol (ARP) in mapping IP addresses to MAC addresses within a local network.
- Configure and test Point-to-Point Protocol (PPP) with authentication methods like CHAP to secure communication over serial links.

Findings:

- **ARP Functionality:**
The experiment demonstrated that ARP plays a vital role in allowing devices to communicate over a LAN by translating IP addresses into MAC addresses. This translation is essential for packet delivery within the same network.
- **PPP with CHAP Authentication:**
The PPP configuration using CHAP (Challenge Handshake Authentication Protocol) ensured a more secure authentication process between two routers. The CHAP method involves a three-way handshake and periodically verifies the identity of the peer.

Problems Faced:

1. Authentication Failure:
Initially, PPP authentication using CHAP failed because the username and password were mismatched between the two routers.
2. Missing ARP Packets:
ARP requests and replies were not visible during packet capture when the network cable was disconnected, causing devices to fail in resolving MAC addresses.

Solutions Implemented:

1. Correcting Authentication Configuration:
The username and password were rechecked and updated to ensure both routers had matching credentials for CHAP authentication.
2. Enabling Interface:
The serial interface was reactivated using the ****no shutdown**** command, restoring connectivity and allowing ARP packets to be exchanged successfully.